Public data, useful test selection, and the remaining evidence gap

A source-intake update and an exploratory rotating-bending transfer pilot

Abstract

New publisher and thesis sources yielded 56 exact reported 316L fatigue lives, 16 separately resolved approximate plot markers, and 13 fitted fatigue-strength parameter rows. These measurement types are retained separately. None supplies a second fully mapped specimen-to-build campaign. An additional rotating-bending replay uses 67 previously collected source outcomes, including a reused 25-point reference curve. Across the same held-out queries, three high-stress tests reduced truncated support cycles by 59.0% at 25 C. The hierarchical q10 life ratio to random selection was 1.450, with observed violation increasing from 0.71% to 7.43%. The random-fatigue-limit ratio was 0.879. At 450 C the hierarchical ratio was 0.761. The evidence shows model- and condition-dependent cost, life and risk tradeoffs; it establishes neither a universally superior policy nor population reliability.

New public evidence and its permitted use

Source	Collected facts and qualification
Zhou / Feng PINN family	44 exact Nf rows from the publisher XLSX, byte-identical to author GitHub data. Three batches are reported but row membership is absent. SEM defect descriptors are measured after fracture. Original stress/range discrepancies retained. Catalog only; zero acquisition/reliability fit rows.
Wietecha-Reiman et al. 2025	12 exact Table 2 lives at three build angles, strain control, 0.25 Hz, HIP condition. Cycles retained without doubling to reversals. No laboratory specimen/build/lot identities; separate strain-controlled panel.
Faue 2022	16 resolved AM markers from a 22-test, one-build report. Approximate lifetimes have pixel intervals; overlap multiplicity, R, control, frequency and temperature remain unresolved. Candidate only.
Jun Zhang 2025 thesis	13 source-fitted strength parameter rows and non-additive counts. These are not 13 Nf observations. R=-1 load-controlled HCF includes explicit reloads after runout; pointwise first exposure needs separation.
Bartosak / Halamka follow-on	Existing 20 identified specimens and 200 defect rows reannotated: CT is pretest, while supplied SWT uses the Nf/2 cycle. No new physical specimens added.

The 441-row source-reviewed axial training input is unchanged. Exact numerical data are not automatically complete-program reliability data. Source family 108486 / 127098 / 111136 is grouped as related work, not three independent campaigns. Article full texts, author code without an explicit license, and original workbooks are omitted from the redistributed factual packet.

A common-query test of stress selection

The separately frozen protocol precedes v5 predictions but follows source transcription: this is exploratory replay. All 67 reported S-N outcomes belong to one conservative 108491 / 109365 family. Ambient conditions supply 44 points and 450 C conditions 23. The 25 C as-built curve reuses 25 prior points; the other 42 points are not proven unique physical specimens. Publication point locators remain distinct from specimen identities.

Six unchanged censored baselines compare fixed 309- and 441-row populations. Four policies share identical query/pool splits, with ten seeds per condition. One 309-trained hierarchical Fisher planner chooses the same sequential supports for all predictors and populations. Candidate/query outcomes are absent from selector inputs. k3 is primary, k5 supporting and feasible k10 ambient results appendix. No target family enters population training.

The primary 5 million-cycle cap makes min(T,cap) support exposure identifiable. Source runouts survive at least 5 million cycles. The 6.07 million-cycle failure remains a source failure and becomes censored only when revealed under the shorter cap. The 10 million-cycle sensitivity retains the known 5 million-cycle censor evidence and reports cost intervals, not invented exact stops. Rotation frequency is unknown, so cycle cost is not translated into hours or money.

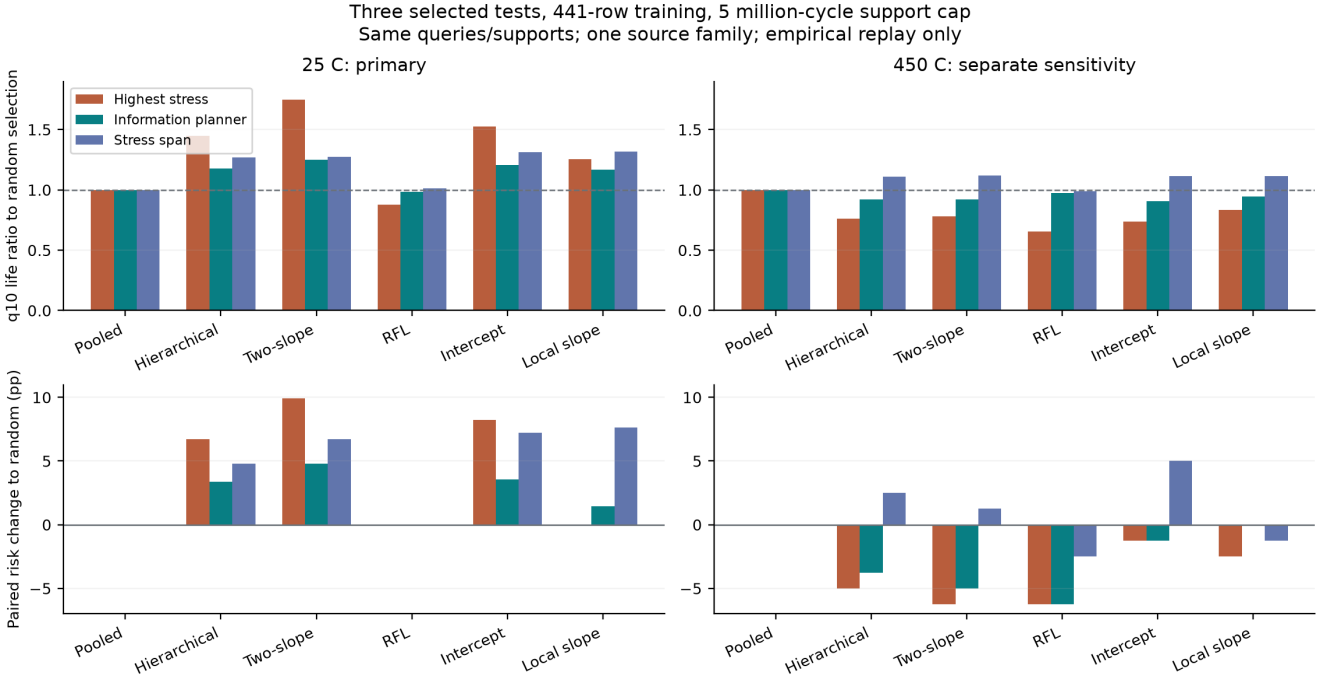


Figure 1. Life ratios and paired risk changes to random at k3. 441-row training, primary support cap. Repeated splits are not independent experiments; confidence intervals for new programs are unavailable. The active planner is shared and is not optimized separately for RFL.

Results and what would count as a contribution

Model	25 C life ratio	25 C risk: random / high	450 C life ratio
Pooled	1.000	0.00% / 0.00%	1.000
Hierarchical	1.450	0.71% / 7.43%	0.761
Two-slope	1.747	4.57% / 14.50%	0.782
RFL	0.879	0.00% / 0.00%	0.654
Intercept	1.526	1.21% / 9.43%	0.737
Local slope	1.256	0.00% / 0.00%	0.836

High-stress versus random, k3: support-cycle savings are 59.0% at 25 C and 77.9% at 450 C, common to every predictor because supports are shared. RFL life ratios are 0.879 and 0.654. The highest-stress strategy can increase useful life for one predictor while reducing it for another. At 450 C it usually buys cost and risk reductions at shorter predicted life. k5, k10, all policies, both populations and both caps remain in the numerical evidence.

Quality control

Independent parent arithmetic checks reconstruct source endpoints, all selected support exposures, common query/support sets, 6,048 absolute summary cells and cost-savings intervals across both caps. Maximum absolute mean difference is 2.23e-16. All eight distinct population model fits report convergence. Matched-intercept k0 predictions correctly use the hierarchical reference; stale last-fit shift fields in its k0 diagnostic objects are excluded from numerical convergence interpretation. The frozen predictive code and reported science are unchanged. Fresh replay and independent paired-contrast receipts accompany the packet.

Claim boundary

This is an auditable public-data benchmark and a counterexample to treating cheap high-stress tests or a single information planner as universally better. It is not yet an established new acquisition method, a second mapped independent campaign, exact reproduction of Centola's 518 records, or superiority to that paper's BNN. Empirical q10 violations from one source family cannot certify R90C90. The original strict criterion remains unresolved: reproducible useful-life improvement against strong baselines at acceptable risk on genuinely independent campaigns.

Next scientific decision

The immediate testable direction is to learn when adaptation should prioritize intercept correction versus fatigue-limit uncertainty, and evaluate a risk/cycle/utility decision rule against the same simple high-stress and local-fit controls. Such a method would require a new outcome-blind protocol and fresh source families; the present results cannot be reused as unlabelled confirmation. If the advantage does not survive independent families, retain the benchmark and reject the method claim.

Primary sources

New numerical PINN source - 10.1016/j.engfracmech.2025.111136

Strain-controlled Table 2 - 10.1007/s10853-025-11259-w

Faue source - 10.1016/j.mfglet.2022.07.083

Bremen thesis - 10.26092/elib/4986

RBF table - 10.1016/j.ijfatigue.2025.109365

Reused RBF reference - 10.1016/j.ijfatigue.2024.108491

Original probabilistic ML study - 10.1016/j.engfailanal.2024.109081